

Abdelrahman Baba

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EDUCATION

University of Waterloo, Waterloo, Canada — Jan. 2023 - Present

Doctor of Philosophy (PhD) in Computer Science

Advisor: Prof. Samer Al-Kiswany

Research Area: Distributed Systems

GPA: 93/100 (A+)

An-Najah National University, Palestine — Sep. 2018 - Dec. 2022

Bachelor of Science (B.Sc.) in Computer Engineering

Advisor: Prof. Ashraf Armoush

GPA: 3.85/4.00 (Excellent)

EXPERIENCE

University of Waterloo

Jan. 2023 - Present

Graduate Research Assistant (RA) and Teaching Assistant (TA)

- Supervisor: Prof. Samer Al-Kiswany.
- My research focuses on enhancing data deduplication methods for encrypted data systems to improve security and efficiency.
- Developing advanced content-defined chunking techniques aimed at reducing data redundancy in large-scale storage systems.
- Investigating machine learning approaches to optimize data storage processes and minimize resource consumption.
- I have worked as a teaching assistant for several courses, including Algorithm Design, Operating Systems, Database Management, and Distributed Systems.

EXALT Technologies Ltd.

Sep. 2021 - Dec. 2022

Design Verification Engineer

- Proficient in Universal Verification Methodology (UVM).
- Implemented and optimized automated verification scripts enhancing test coverage by 25%.
- Mentored and trained 3 junior engineers and interns in advanced verification methodologies, particularly focusing on UVM, system-level testing, and automation practices.

PUBLICATIONS

- VectorCDC: Accelerating Data Deduplication with SSE/AVX Instructions, *FAST'25*
- SeqCDC: Hashless Content-Defined Chunking for Data Deduplication, *Middleware'24*
- The Impact of Low-Entropy on Chunking Techniques for Data Deduplication, *CLOUD'24*
- DedupBench: A Benchmarking Tool for Data Chunking Techniques, *CCECE 2023*.

PATENTS

- Systems and methods for executing jump-based content-defined data chunking
Published via USPTO (Pending).
- System and Method of Deduplication for Encrypted Content.
Published via USPTO (Pending).

AWARDS

- Acronis Best Patent Award for 2023.
- University of Waterloo Graduate Student Research Dissemination Award 2023.
- University of Waterloo International Doctoral Student Award (IDSA).

RESEARCH PROJECTS

Serverless Resource Optimization

- Comprehensive analysis of real-world serverless production traces to identify serverless workload characteristics.
- Proposed an analytical approach to optimize the pool of pre-warmed resources in serverless environments.
- The proposed approach increases resource utilization, reducing running costs.

Resemblance-Detection for Post-Deduplication Delta Compression

- Delta compression enhances the compression ratio in deduplication systems.
- Analyzed current resemblance detection techniques which utilize exact sketch-matching to identify similar data blocks.
- Designed a system architecture for advanced resemblance detection using fuzzy hashing and nearest neighbor searches for summary digest.

Content-Defined Chunking (Middleware'24, FAST'25)

- Content-defined chunking divides data into variable-size chunks to improve the compression ratio.
- Content-defined chunking is commonly used in file synchronization and data deduplication.
- Developed a novel chunking technique that utilizes speculative jumps, achieving a speed that is 4-10x than the current state-of-the-art methods.

A Study of SmartNICs Applications

- Studied different architectures and design paradigms of SmartNICs.
- Conducted a comprehensive study on SmartNICs applications to identify and evaluate their use cases.
- Outlined future research directions and potential areas for improvement in SmartNIC technologies.